Take Home Exam 2

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Short Report in a concise 2-page report

1. **Introduction**: State the objective and describe the dataset.

The objective of the dataset is to help us analyze the relationship between health scores and various potential personal factors, like Age, BMI, Exercise Hours, Smoking Status, and Income. The dataset is a csv file which includes 6 columns with the variables mentioned just now. Here, we use the dataset containing these variables to explore correlations, calculate prediction intervals and predict health scores with simple and multiple linear regression analysis and bootstrap techniques.

2. **Exploratory Data Analysis**

In Exploratory Data Analysis, the question mainly focus on numerical variables. To begin with, the statistics of 5 relevant numerical variables are summarized, including mean, median, standard deviation, lower quartile, upper quartile, minimum and maximum. The detailed value of these statistics data will show below. At the same time, these variables are required to be visualized in this part. According to the Histogram of Numerical Variables, it looks that BMI, Health_Score and Income are spread in the normal distribution. The Boxplot of Numerical Variables shows that BMI and Imcome have outliers while Age and Exercise_Hours look more dispersed. The Visualizations of the Response Variable against Numerical Predictor Variables indicates that Health_Scores has positive relationships with Exercise_Hours and Income while negative relationships with Age and BMI. The boxplot below about Smoking_Status reveals that the median and lower and upper quartiles of Health_Scores of smokers are lower than that of non-smokers. As for the initial observations, based on the data manipulating process, BMI and Income variables both have 2 outliers. Age_Skewness, BMI_Skewness, and Exercise_Hours_Skewness show left skewness while the other two numerical variables show right skewness. However, the magnitude of their skewness are all small. None of the columns have missing values.

3. **Simple Linear Regression**

In Simple Linear Regression, according to the 4 correlation coefficients, the correlation coefficient between Income and Health_Score has the largest magnitude of 0.6687468. This means Income is the most reasonable predictor. Then, I can make the linear regression model with lm(). The estimated regression equation is Health_Score = 59.5210944765007 + 0.000477127394928402 * Income. When the Income is 0, the estimated Health_Score is about 59.521. For each additional 1 unit increase in Income, Health_Score increases by about 0.000477. For simple linear regression models, r_squared = r ^ 2. Here, the R-squared value is about 0.4472222. In the last part of Simple Linear Regression, 2 graphs are plotted. The first graph is the residual plot. The points all look evenly separated without any patterns. Based on the second graph, the residuals spread in normal distribution because the bars holistically form a bell shape, which is the same as normal distribution.

4. **Bootstrapping for Interference**

In Bootstrapping for Interference, the distribution of the bootstrapped slopes is shown below. It roughly forms a bell shape which makes it look like normal distribution. We are about 95% confident that the

Health_Score tends to increase by between 0.0004143684 and 0.0005404859 units every one unit increase in Income on average. At the same time, the predicted Health_Score here for the question is 71.44928 based on the linear regression model. According to the bootstrapping process, the 95% prediction interval for the response variable is (69.49780, 73.35643). The predict value is in the 95% prediction interval, which is acceptable.

5. **Multiple Linear Regression**

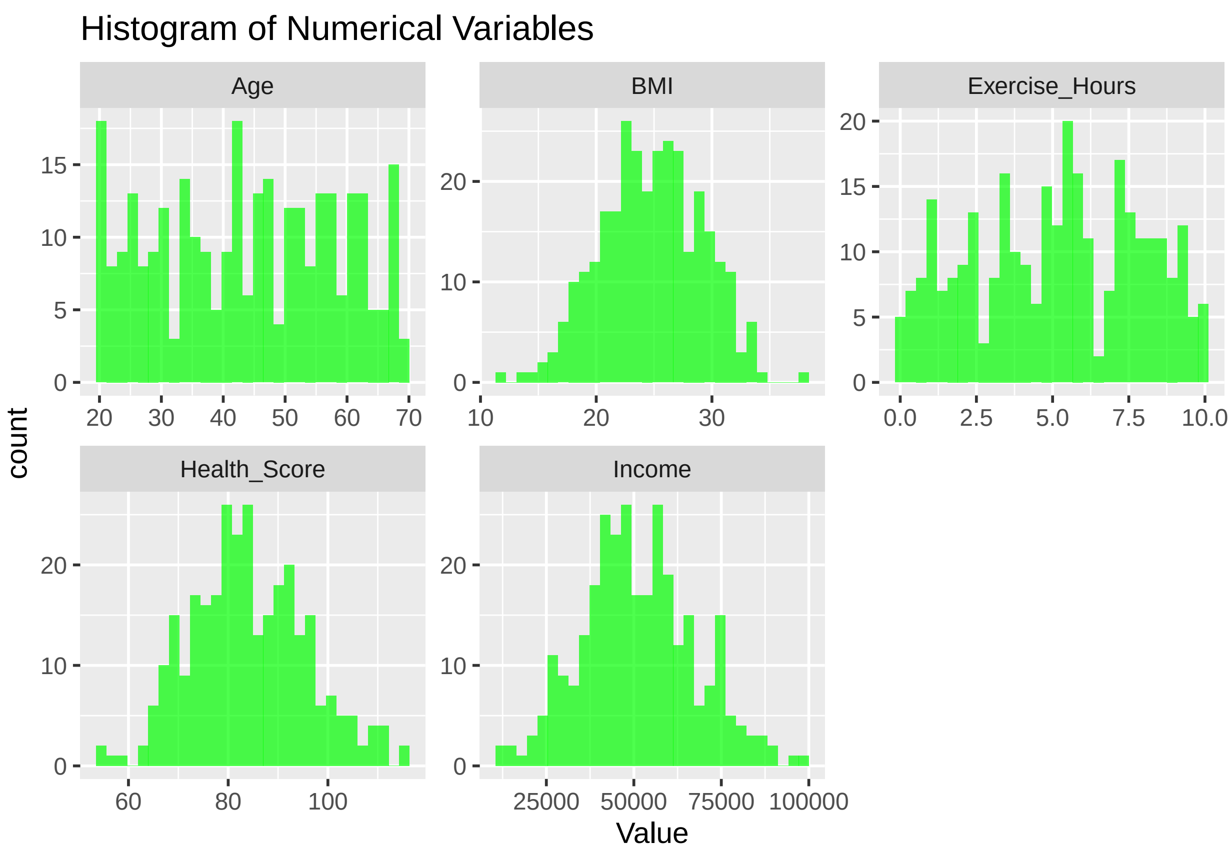
In Multiple Linear Regression, the equation of the multiple linear regression model comes from plugging out the coefficients. Based on the model, the equation is Health_Score = 80.0935965698827 + -0.306567220507554 * Age + -0.556570644809524 * BMI + 1.47363461678577 * Exercise_Hours + 0.00046109957830263 * Income. The estimated intercept is 80.0935965699. When all the four variables are 0, the estimated Health_Score is about 80.094. The estimated slope coefficient is -0.306567220507554 (Age), -0.556570644809524 (BMI), 1.47363461678577 (Exercise_Hours), and 0.00046109957830263 (Income). For each additional year of Age, Health_Score decreases by about 0.307 points. For each 1-unit increase in BMI, Health_Score decreases by about 0.557 points. For each additional hour of exercise, Health_Score increases by about 1.474 points. For each additional 1 unit increase in income, Health_Score increases by about 0.000461. According to the comparison between simple and multiple linear regression analysis, the adjusted R-squared of the simple linear regression is 0.4453673 while that of the multiple linear regression is 0.7619176. The multiple linear regression model has a much larger adjusted R-squared. This mean the multiple linear regression model is more accurate. Last but not least, the multiple linear regression predicted Health_Score is 77.24903 and its prediction interval is (75.06579, 79.32674). The simple linear regression model has a predicted Health_Score of 71.44928 and a prediction interval of (69.49780, 73.35643). Compared with the simple linear regression model, the multiple linear regression model has a larger predicted Health_Score and a larger prediction interval in values.

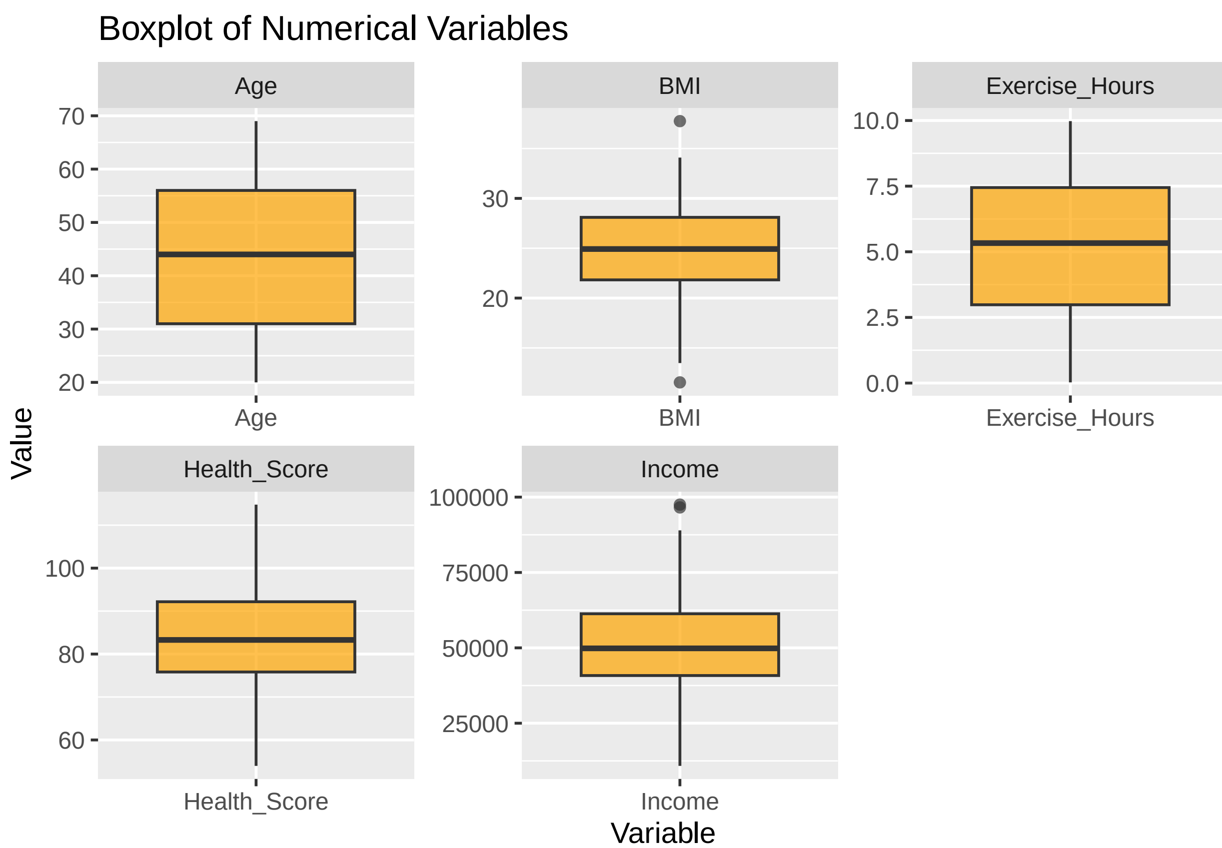
6. **Conclusions**: Summarize major results from EDA, regression analysis, and bootstrapping. Interpret results in the context of the data, mention limitations, and suggest next steps.

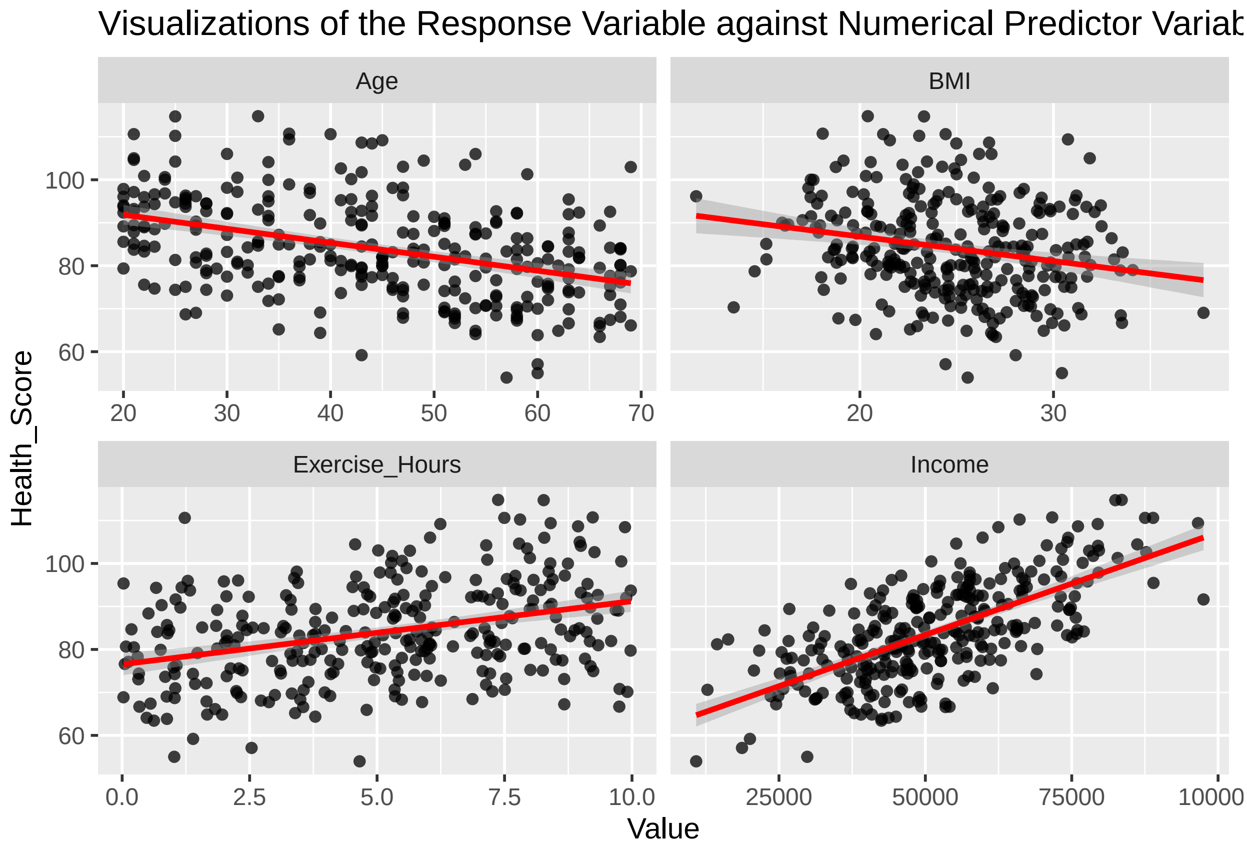
In conclusion, the 5 variables all seem to be factors of Health_Scores. Typically, Health_Scores has positive relationships with Exercise_Hours and Income while negative relationships with Smoking_Status, Age and BMI. Among those factors, Income has the highest level of correlation. With the help of bootstrapping technique, the prediction value of the data specified in the question is acceptable for simple linear regression case. The multiple linear regression model looks more reliable and accurate because it has a higher adjust R-squared than the simple linear regression model. The equations have already been shown in the previous explanation parts. There are some assumptions and limitations of the regression analysis. The relationship between predictors and Health_Score should be linear. Residuals should not be correlated. Residuals should have equal variance. "Residuals should follow a normal distribution. Predictors should not be highly correlated. If the residual plot shows a pattern, we need to check some other better regression models. If residuals are not normal, we need to remove outliers or using a robust regression model.

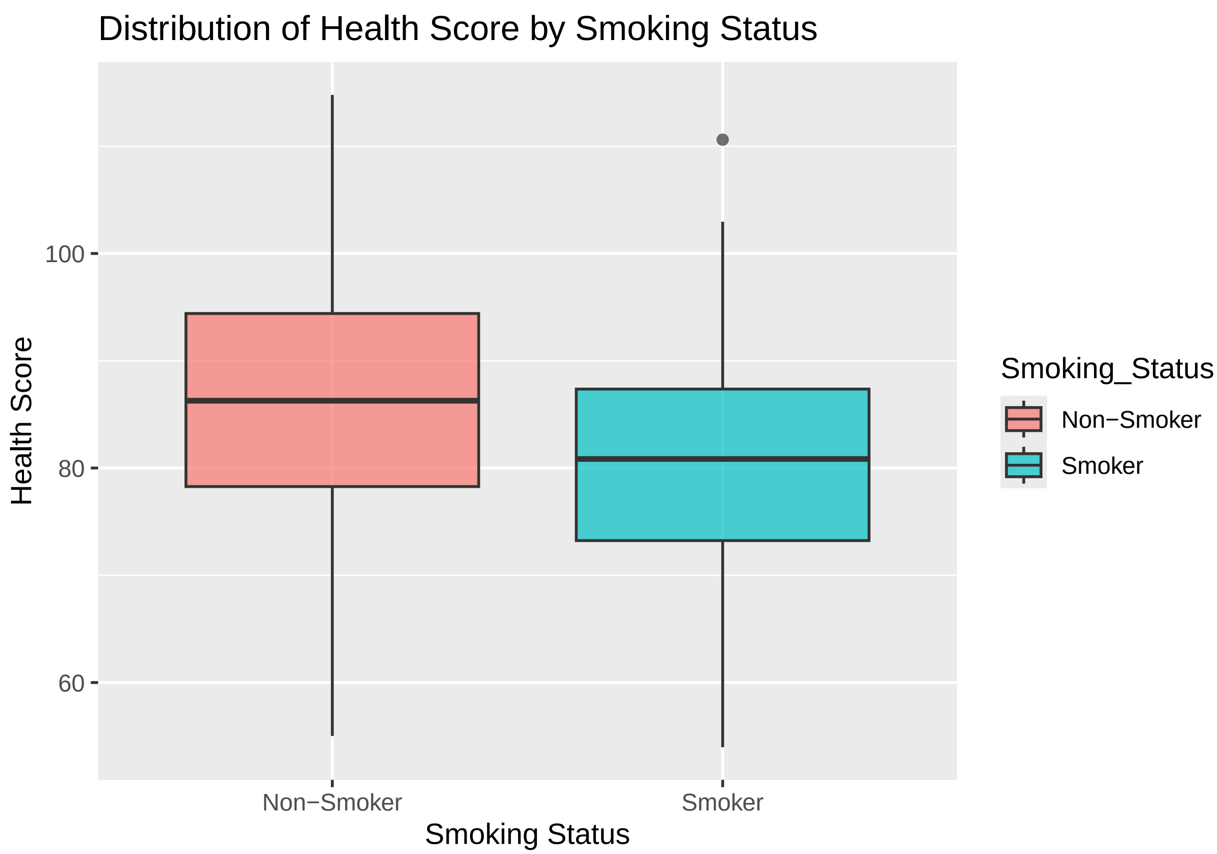
7. **Appendix**: Include any plots or tables with clear labels (not included in page limit).

```
## # A tibble: 1 x 35
##   Age_Mean Age_Median Age_SD Age_Min Age_Q1 Age_Q3 Age_Max BMI_Mean BMI_Median
##      <dbl>      <dbl>  <dbl>   <dbl>  <dbl>  <dbl>   <dbl>    <dbl>      <dbl>
## 1     44.0         44   14.5      20     31     56      69     24.8       24.9
## # i 26 more variables: BMI_SD <dbl>, BMI_Min <dbl>, BMI_Q1 <dbl>, BMI_Q3 <dbl>,
## #   BMI_Max <dbl>, Exercise_Hours_Mean <dbl>, Exercise_Hours_Median <dbl>,
## #   Exercise_Hours_SD <dbl>, Exercise_Hours_Min <dbl>, Exercise_Hours_Q1 <dbl>,
## #   Exercise_Hours_Q3 <dbl>, Exercise_Hours_Max <dbl>, Income_Mean <dbl>,
## #   Income_Median <dbl>, Income_SD <dbl>, Income_Min <dbl>, Income_Q1 <dbl>,
## #   Income_Q3 <dbl>, Income_Max <dbl>, Health_Score_Mean <dbl>,
## #   Health_Score_Median <dbl>, Health_Score_SD <dbl>, ...
```


Histogram of Numerical Variables
Age
BMI
Exercise_Hours
Health_Score
Income
count
Value


Boxplot of Numerical Variables
Age
70
60
50
40
30
20
Age
BMI
30
20
BMI
Exercise_Hours
10.0
7.5
5.0
2.5
0.0
Exercise_Hours
Value
Health_Score
100
80
60
Health_Score
Income
100000
75000
50000
25000
Income
Variable

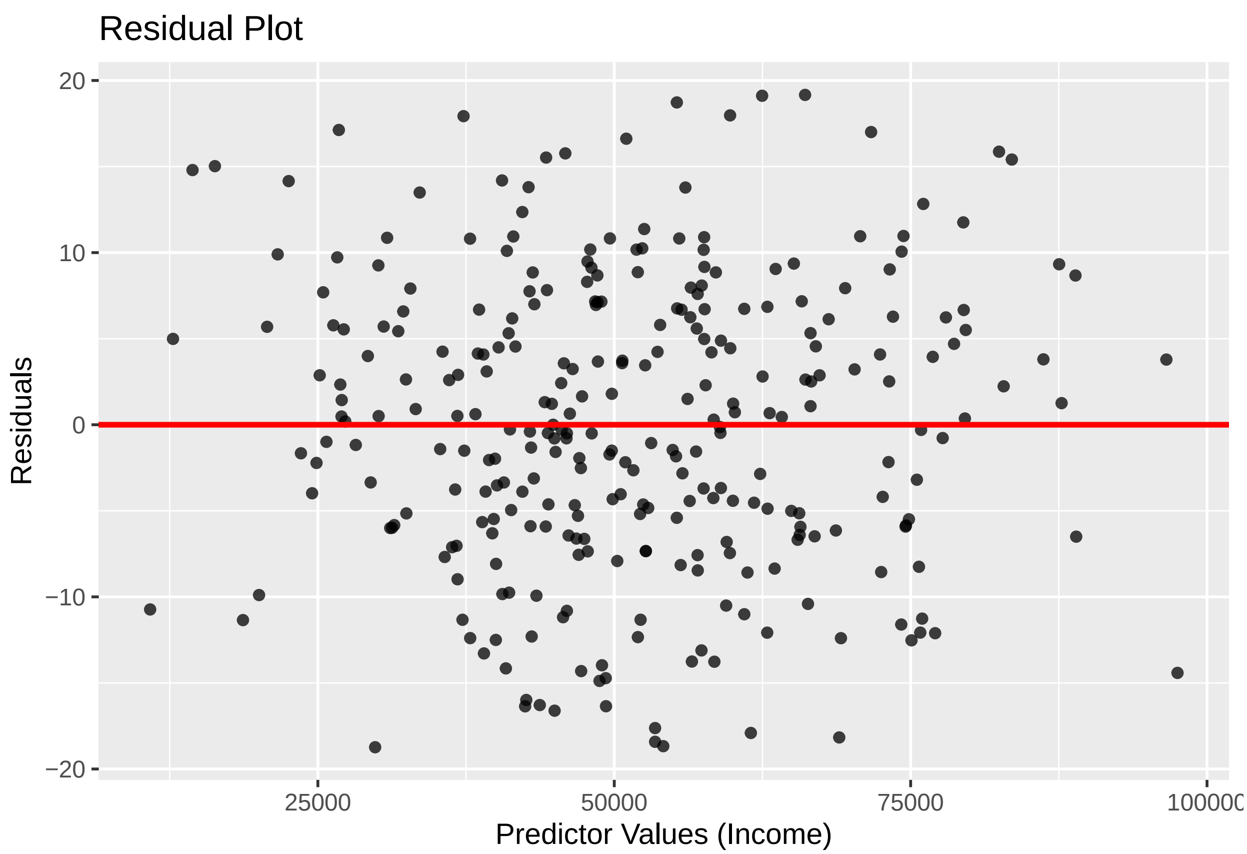

Visualizations of the Response Variable against Numerical Predictor Variat
Age
BMI
Exercise_Hours
Income
Health_Score
Value
100
80
60
20
30
40
50
60
70
0.0
2.5
5.0
7.5
10.0
25000
50000
75000
10000

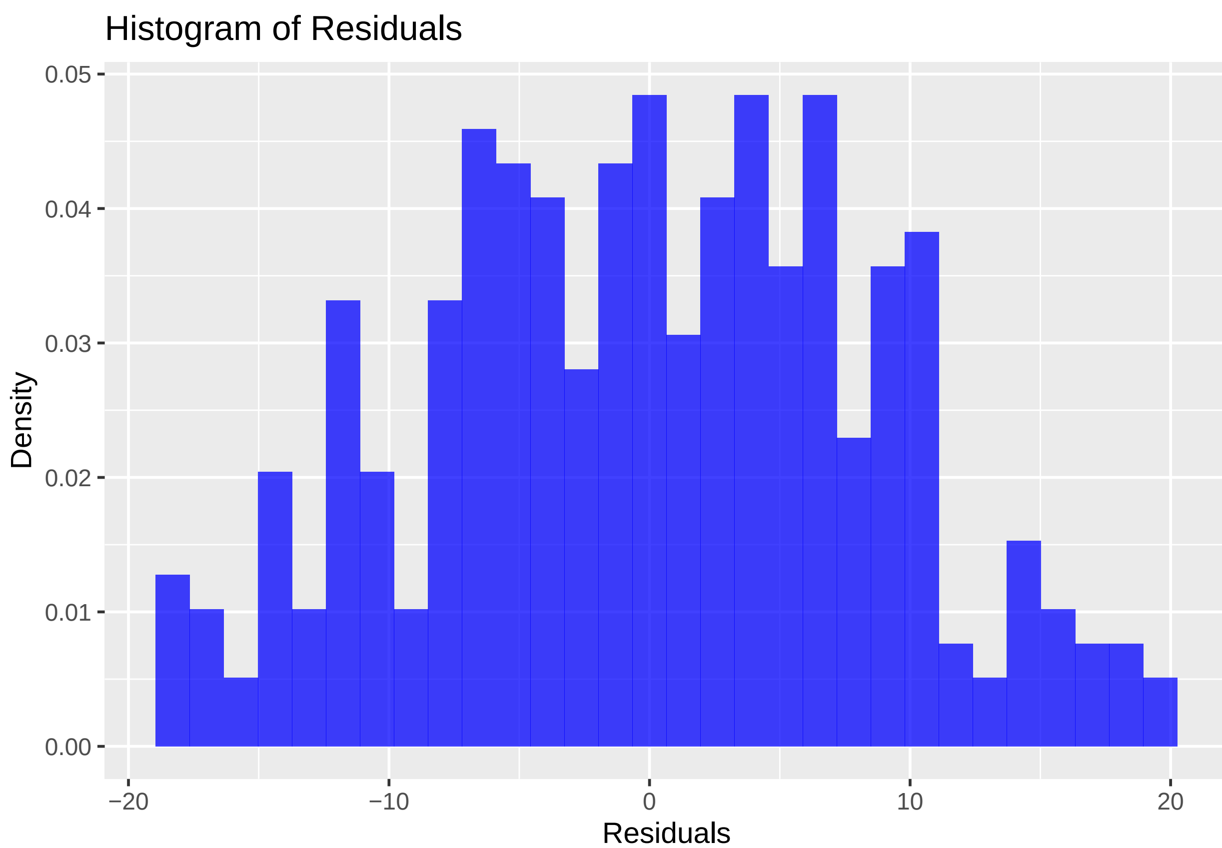


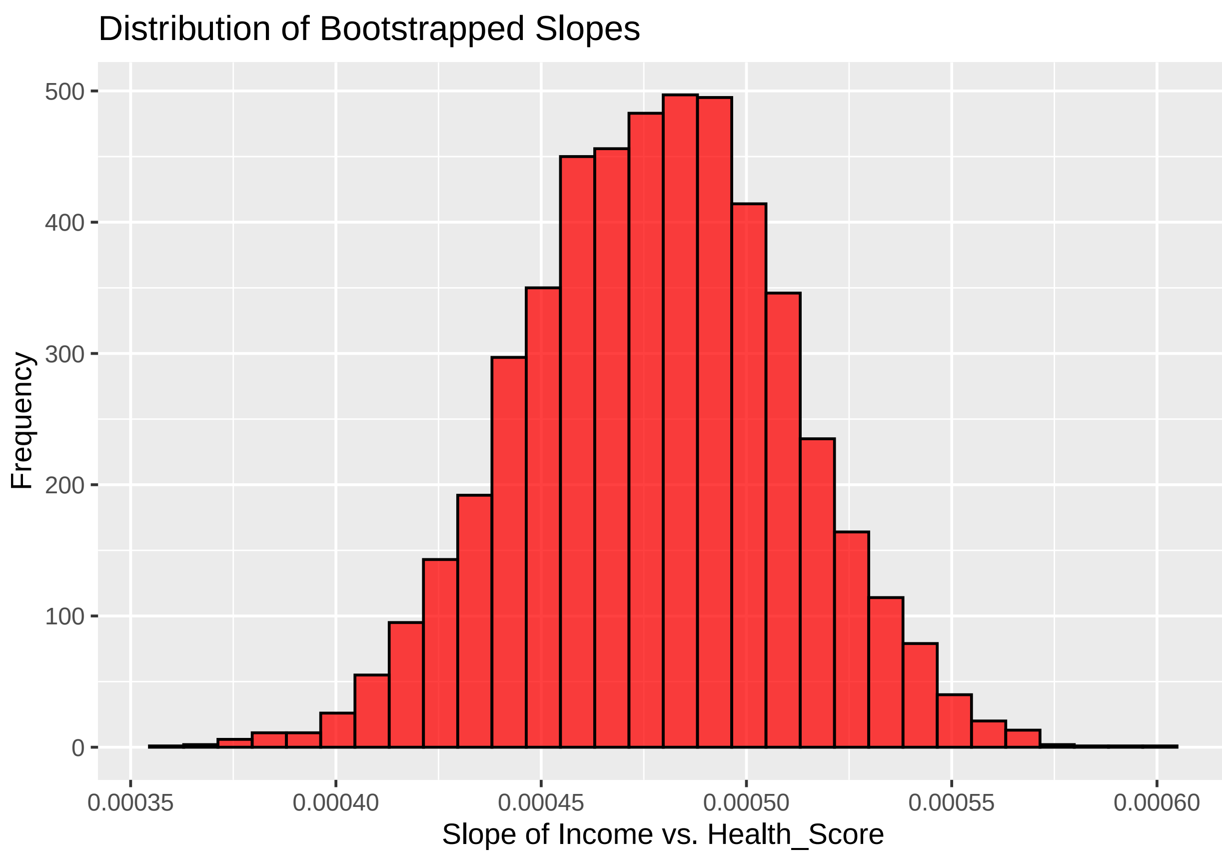

```
## # A tibble: 1 x 5
##     Age   BMI Exercise_Hours Income Health_Score
##   <int> <int>          <int>  <int>        <int>
## 1     0     2              0      2            0
```

```
## # A tibble: 1 x 5
##   Age_Skewness BMI_Skewness Exercise_Hours_Skewness Income_Skewness
##          <dbl>        <dbl>                   <dbl>           <dbl>
## 1      -0.0358      -0.0805                 -0.0999           0.190
## # i 1 more variable: Health_Score_Skewness <dbl>
```

```
## # A tibble: 1 x 6
##   Missing_Age Missing_BMI Missing_Exercise_Hours Missing_Smoking_Status
##         <int>       <int>                  <int>                  <int>
## 1           0           0                      0                      0
## # i 2 more variables: Missing_Income <int>, Missing_Health_Score <int>
```


Residual Plot
20
10
0
−10
−20
Residuals
25000
50000
75000
100000
Predictor Values (Income)


Histogram of Residuals
Density
0.05
0.04
0.03
0.02
0.01
0.00
−20
−10
0
10
20
Residuals




```
## # A tibble: 2 x 2
##   Model                          Adjusted_R2
##   <chr>                                <dbl>
## 1 Simple Linear Regression             0.445
## 2 Multiple Linear Regression           0.762
```

```
## # A tibble: 2 x 4
##   Model      Predicted_Health_Score `95% PI (Lower Bound)` `95% PI (Upper Bound)`
##   <chr>                       <dbl>                  <dbl>                  <dbl>
## 1 Simple R~                    71.4                   69.5                   73.4
## 2 Multiple~                    77.2                   75.1                   79.3
```

8. **R Appendix**: Include any R codes you used for the report. Do not show your code in the body of the report (1-6). (not included in page limit).

```r
##### IMPORT THE DATASET
library(tidyverse)
# import the datasets
health <- read_csv("health_dataset.csv")
```

```
## Rows: 300 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): Smoking_Status
## dbl (5): Age, BMI, Exercise_Hours, Income, Health_Score
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# briefly know the structures of the three csv files with head()
health %>% head()
```

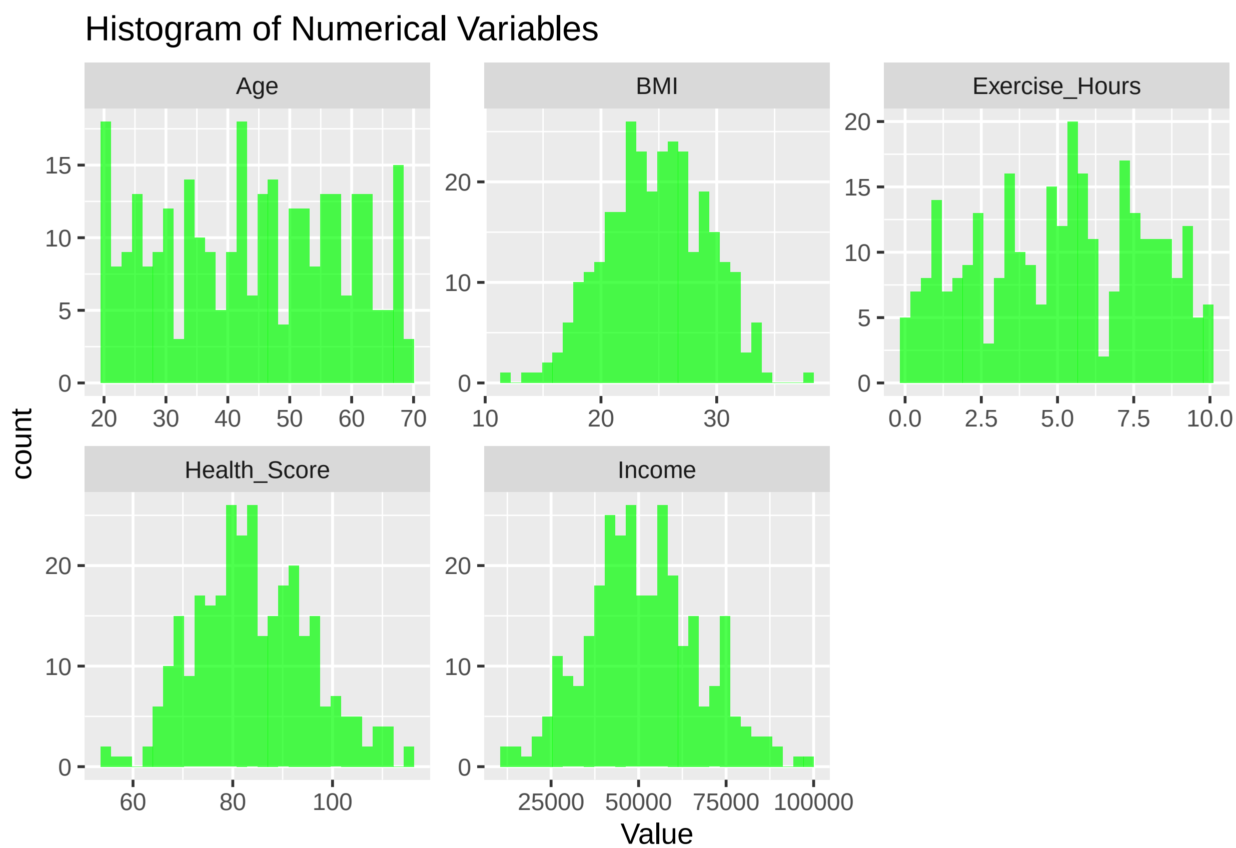
```
## # A tibble: 6 x 6
##     Age   BMI Exercise_Hours Smoking_Status Income Health_Score
##   <dbl> <dbl>          <dbl> <chr>           <dbl>        <dbl>
## 1    58  27.6           6.97 Smoker         52179.         79.2
## 2    48  29.4           4.12 Smoker         75704.         87.4
## 3    34  17.6           8.74 Smoker         65152.        100. 
## 4    62  29.7           5.15 Non-Smoker     47056.         80.0
## 5    27  21.1           9.73 Smoker         33588.         89.0
## 6    40  20.6           6.02 Non-Smoker     14426.         81.2
```

```r
##### HERE IS QUESTION 1 OF PART 1
health %>%
  # need to find the columns with numerical data first
  select(where(is.numeric)) %>%
  # use summarise() function to show these statistical properties
  summarise(
    # apply multiple columns inside the summarise() with across()
    across(
      # select all columns with everything()
      everything(),
      # apply multiple function inside across() using list()
      list(
        # ~ means anonymous function, applying to multiple functions
        Mean = ~mean(.x, na.rm = TRUE), # mean
        Median = ~median(.x, na.rm = TRUE), # median
        SD = ~sd(.x, na.rm = TRUE), # standard deviation
        Min = ~min(.x, na.rm = TRUE), # minimum
        Q1 = ~quantile(.x, 0.25, na.rm = TRUE), # lower quantile
        Q3 = ~quantile(.x, 0.75, na.rm = TRUE), # upper quantile
        Max = ~max(.x, na.rm = TRUE) # maximum
      )
    )
  )
```

```
## # A tibble: 1 x 35
##   Age_Mean Age_Median Age_SD Age_Min Age_Q1 Age_Q3 Age_Max BMI_Mean BMI_Median
##      <dbl>      <dbl>  <dbl>   <dbl>  <dbl>  <dbl>   <dbl>    <dbl>      <dbl>
## 1     44.0         44   14.5      20     31     56      69     24.8       24.9
## # i 26 more variables: BMI_SD <dbl>, BMI_Min <dbl>, BMI_Q1 <dbl>, BMI_Q3 <dbl>,
## #   BMI_Max <dbl>, Exercise_Hours_Mean <dbl>, Exercise_Hours_Median <dbl>,
## #   Exercise_Hours_SD <dbl>, Exercise_Hours_Min <dbl>, Exercise_Hours_Q1 <dbl>,
## #   Exercise_Hours_Q3 <dbl>, Exercise_Hours_Max <dbl>, Income_Mean <dbl>,
## #   Income_Median <dbl>, Income_SD <dbl>, Income_Min <dbl>, Income_Q1 <dbl>,
## #   Income_Q3 <dbl>, Income_Max <dbl>, Health_Score_Mean <dbl>,
## #   Health_Score_Median <dbl>, Health_Score_SD <dbl>, ...
```

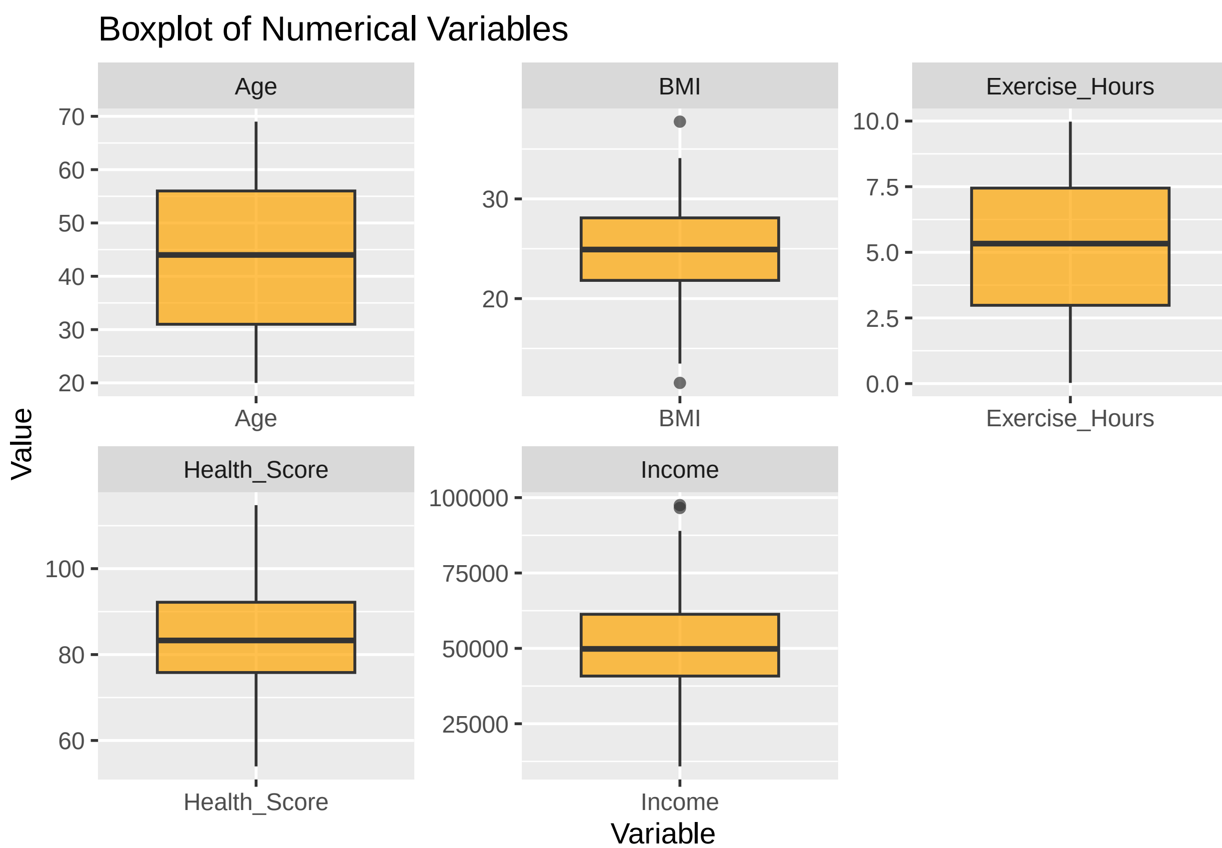
```r
### **interpretation:** The tibble here shows the summary statistics of relevant numerical
### variables, including mean, median, standard deviation, minimum, lower and upper quartle,
### and maximum.
```

```r
##### HERE IS QUESTION 2 OF PART 1
# make a histogram of numerical variables
health %>%
  # use pivot_longer() to make the tibble into a longer form
  pivot_longer(cols = where(is.numeric), names_to = "Variable", values_to = "Value") %>%
  # the x-axes are the values of these variables
  # the y-axes are the frequencies of these values
  ggplot(aes(x = Value)) +
  # set the number of bars, color and degree of transparency
  geom_histogram(bins = 30, fill = "green", alpha = 0.7) +
  # separate into several sub-histograms
  facet_wrap(~ Variable, scales = "free") +
  # set the title
  labs(title = "Histogram of Numerical Variables")
```

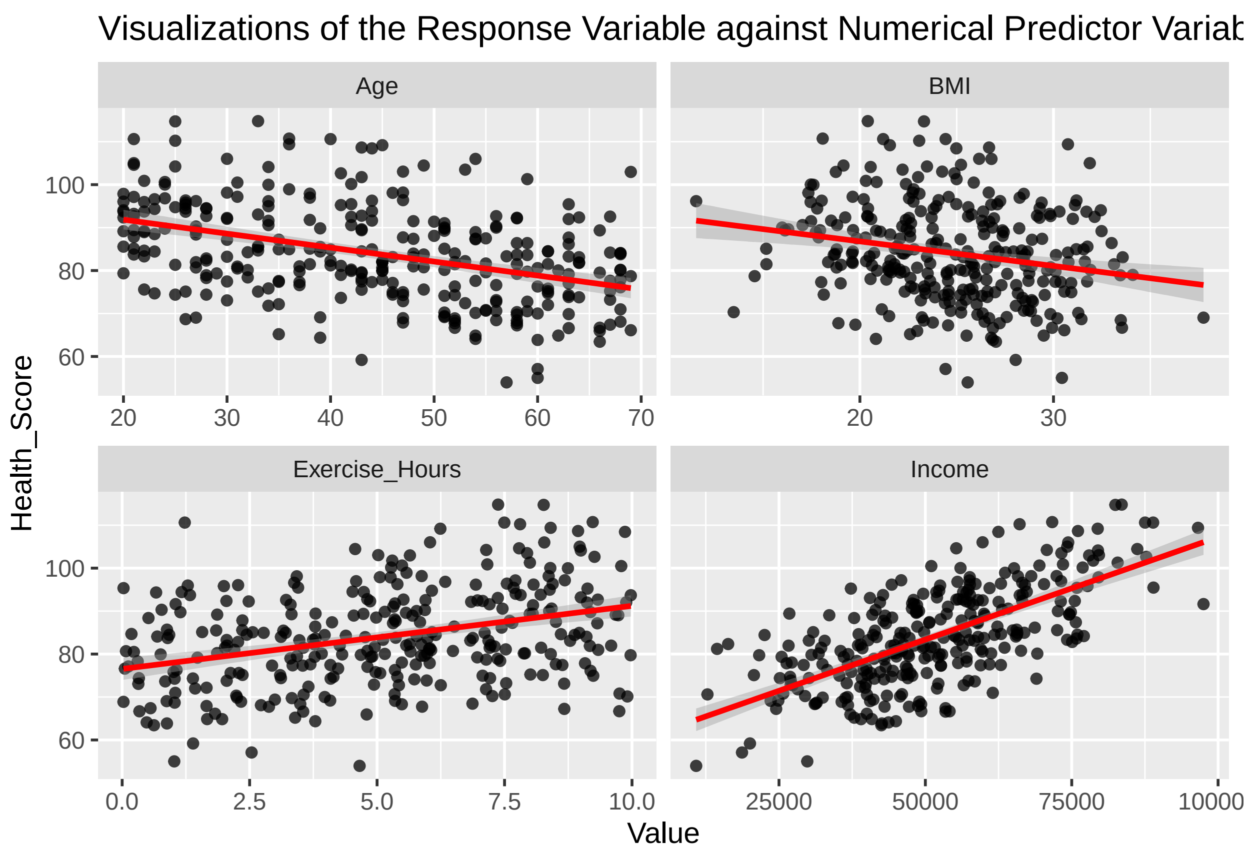



```r
# make a boxplot of numerical variables to show the variation
health %>%
  # use pivot_longer() to make the tibble into a longer form
  pivot_longer(cols = where(is.numeric), names_to = "Variable", values_to = "Value") %>%
  # the x-axes are the names of these variables
  # the y-axes are the values of these variables
  ggplot(aes(x = Variable, y = Value)) +
  # draw the boxplot and the set the degree of transparency
```

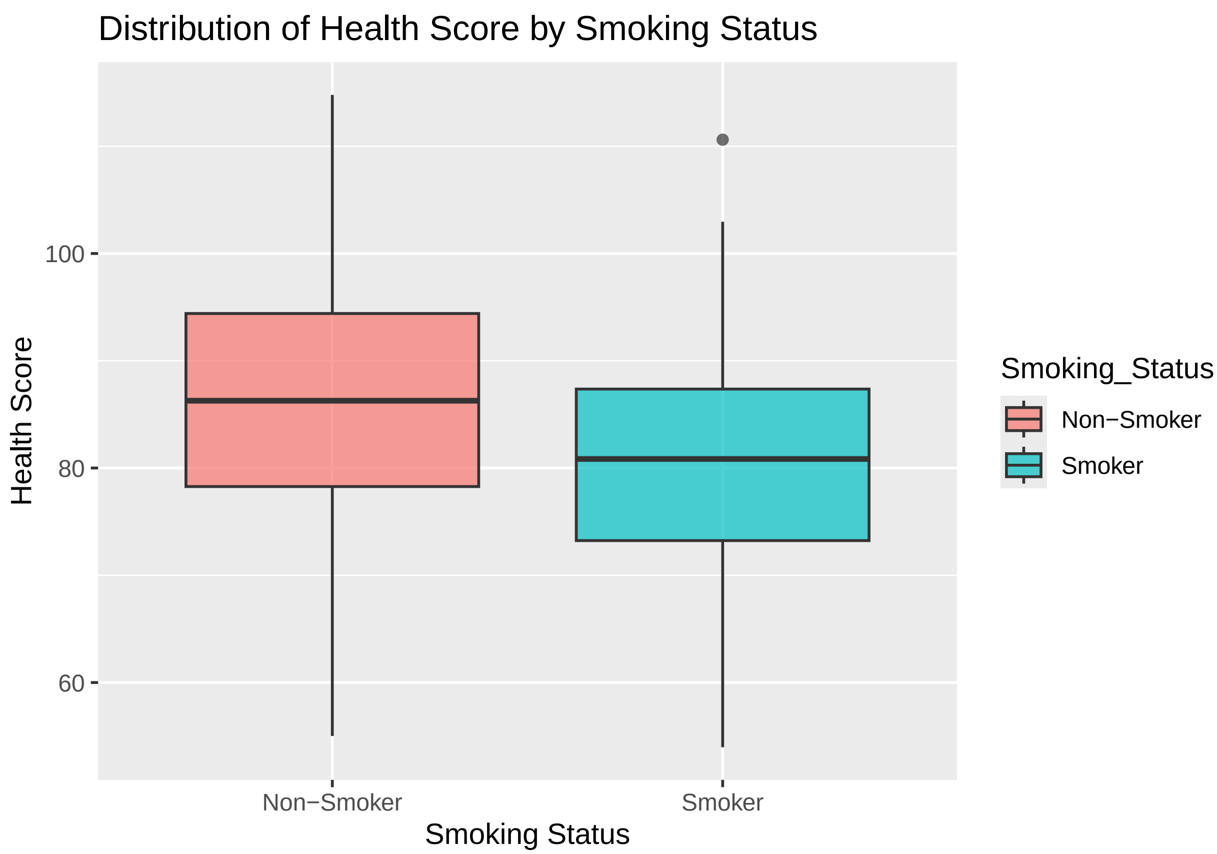
```r
  geom_boxplot(fill = "orange", alpha = 0.7) +
  # separate into several sub-boxplots
  facet_wrap(~ Variable, scales = "free") +
  # set the title
  labs(title = "Boxplot of Numerical Variables")
```




```r
# set the predictors
predictors <- c("Age", "BMI", "Exercise_Hours", "Income")
# set the response
response <- "Health_Score"

# make a scatter plot of numerical variables to show the correlation and trend
health %>%
  # use pivot_longer() to make the tibble into a longer form
  pivot_longer(cols = all_of(predictors), names_to = "Predictor", values_to = "Value") %>%
  # set the x-axes as different predictors and y-axes as Health_Score
  ggplot(aes(x = Value, y = Health_Score)) +
  # set the degree of transparency
  geom_point(alpha = 0.75) +
  # draw the best fit line
  geom_smooth(method = "lm", formula = y ~ x, color = "red") +
  # separate into several sub-scatter plots
  facet_wrap(~ Predictor, scales = "free_x") +
  # set the title
  labs(title = "Visualizations of the Response Variable against Numerical Predictor Variables")
```




```r
# make a boxplot of categorical variables to show the correlation and trend
health %>%
  # group data by smoker vs. non-smoker
  group_by(Smoking_Status) %>%
  # set the x-axes as Smoking_Status and y-axes as Health_Score
  ggplot(aes(x = Smoking_Status, y = Health_Score, fill = Smoking_Status)) +
  # Box plot to show quartiles, median, and outliers
  geom_boxplot(alpha = 0.7) +
  # set the title, x-axis and y-axis
  labs(title = "Distribution of Health Score by Smoking Status",
       x = "Smoking Status",
       y = "Health Score")
```




```
### **Interpretation:** The first graph is histograms of numerical variables which shows
### the central tendency. The second graph is boxplots of numerical variables which shows
### spread variation. The third graph is the scatter plots of numerical variables vs.
### Health_Score which shows the correlation and relationship. The fourth graph shows the
### relationship between non-numerical variables (smoking status) and Health_Score.





##### HERE IS QUESTION 3 OF PART 1
# check outliers
# define a function to return the number of outliers
detect_outliers <- function(x) {
  # find the lower quartile
  Q1 <- quantile(x, 0.25, na.rm = TRUE)
  # find the upper quartile
  Q3 <- quantile(x, 0.75, na.rm = TRUE)
  # calculate the interquartile range
  IQR_value <- Q3 - Q1
  # count the number of outliers
  num_outliers <- sum(x < (Q1 - 1.5 * IQR_value) | x > (Q3 + 1.5 * IQR_value), na.rm = TRUE)
  # return the number_outliers
  return(num_outliers)
}
```

```r
# apply the detect_outliers to every column
health %>%
  # select the columns which are numerical
  select(where(is.numeric)) %>%
  # summarise the number of ouliers for every numerical column
  summarise(across(everything(), detect_outliers))
```

```
## # A tibble: 1 x 5
##     Age   BMI Exercise_Hours Income Health_Score
##   <int> <int>          <int>  <int>        <int>
## 1     0     2              0      2            0
```

```r
# check the skewness
# define a function to return the skewness
# the formula of skewness is {n/[(n-1)(n-2)]}sigma[(xi-x_bar)^2/sd^3]
skewness <- function(x) {
  # count the total number of observations
  n <- sum(!is.na(x))
  # calculate the mean
  mean_x <- mean(x, na.rm = TRUE)
  # calculate the variance
  var_x <- var(x, na.rm = TRUE)
  # calculate the adjusted factor
  adj_factor <- n / ((n - 1) * (n - 2))
  # plug the 3 defined variables into the formula of skewness
  skew <- adj_factor * sum((x - mean_x)^3, na.rm = TRUE) / (var_x^(3/2))
  # return the value of skewness
  return(skew)
}
health %>%
  # select the columns which are numerical
  select(where(is.numeric)) %>%
  # use summarise() function to show these statistical properties
  # apply multiple columns inside the summarise() with across()
  # select all columns with everything()
  summarise(across(everything(), list(
    Mean = ~mean(.x, na.rm = TRUE),
    Median = ~median(.x, na.rm = TRUE),
    SD = ~sd(.x, na.rm = TRUE),
    Min = ~min(.x, na.rm = TRUE),
    Q1 = ~quantile(.x, 0.25, na.rm = TRUE),
    Q3 = ~quantile(.x, 0.75, na.rm = TRUE),
    Max = ~max(.x, na.rm = TRUE),
    # calculate the skewness for all the numerical columns
    Skewness = ~skewness(.x)
  ))) %>%
  # select the skewness columns
  select(contains("Skewness"))
```

```
## # A tibble: 1 x 5
##   Age_Skewness BMI_Skewness Exercise_Hours_Skewness Income_Skewness
##          <dbl>        <dbl>                   <dbl>           <dbl>
## 1      -0.0358      -0.0805                 -0.0999           0.190
## # i 1 more variable: Health_Score_Skewness <dbl>
```

```r
# check the missing values
health %>%
  # count the number of missing values of every column in the similar way as I described before
  summarise(across(everything(), ~sum(is.na(.)), .names = "Missing_{.col}"))
```

```
## # A tibble: 1 x 6
##   Missing_Age Missing_BMI Missing_Exercise_Hours Missing_Smoking_Status
##         <int>       <int>                  <int>                  <int>
## 1           0           0                      0                      0
## # i 2 more variables: Missing_Income <int>, Missing_Health_Score <int>
```

```r
### **Interpretation:** Based on the data manipulating process, BMI and Income variables
### both have 2 outliers. Age_Skewness, BMI_Skewness, and Exercise_Hours_Skewness show
### left skewness while the other two numerical variables show right skewness. However,
### the maginitude of their skewness are all small. None of the columns have missing values.







##### HERE IS QUESTION 1 OF PART 2
# find most reasonable predictor by comparing their correlation coefficients
r_age <- cor(health$Health_Score, health$Age) # Age vs. Health_Score
r_bmi <- cor(health$Health_Score, health$BMI) # BMI vs. Health_Score
r_exercise <- cor(health$Health_Score, health$Exercise_Hours) # Exercise_Hours vs. Health_Score
r_income <- cor(health$Health_Score, health$Income) # Income vs. Health_Score
# create a vector of the 4 correlation coefficients
cor_vec = c(r_age, r_bmi, r_exercise, r_income)
cor_vec
```

```
## [1] -0.4129672 -0.2179569  0.3542721  0.6687468
```

```r
# Income vs. Health_Score has the largest maginitude of correlation coefficient
# create the linear regression model
lm_model <- lm(Health_Score ~ Income, data = health)
# get the value of intercept
intercept <- coef(lm_model)[1]
# get the value of slope
slope <- coef(lm_model)[2]
# show the linear equation
equation <- paste("Health_Score =", intercept, "+", slope, "* Income")
equation
```

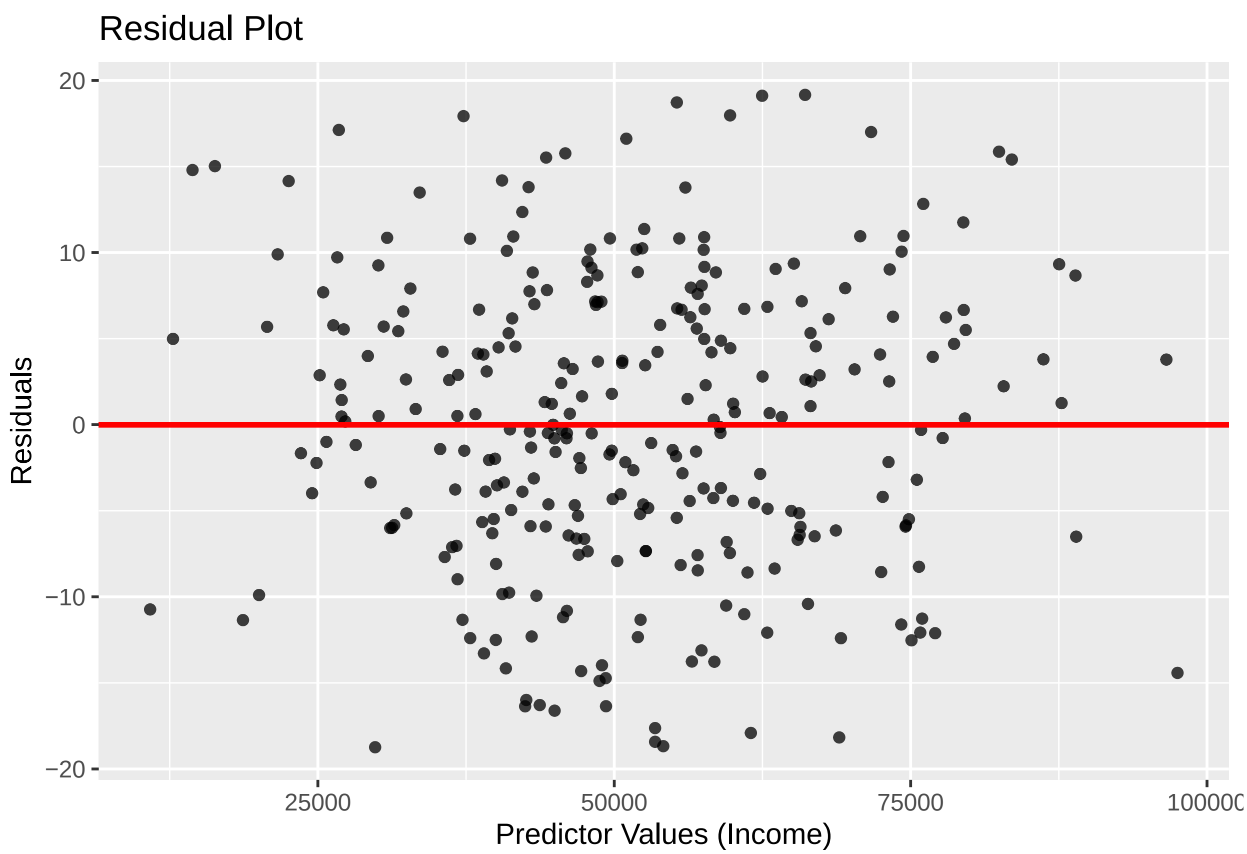
```
## [1] "Health_Score = 59.5210944765007 + 0.000477127394928402 * Income"
```

```r
### **Interpretation:** According to the 4 correlation coefficients, the correlation
### coefficient between Income and Health_Score has the largest magnitude. This means
### Income is the most reasonable predictor. Then, I can make the linear regression model
### with lm().





##### HERE IS QUESTION 2 OF PART 2
```

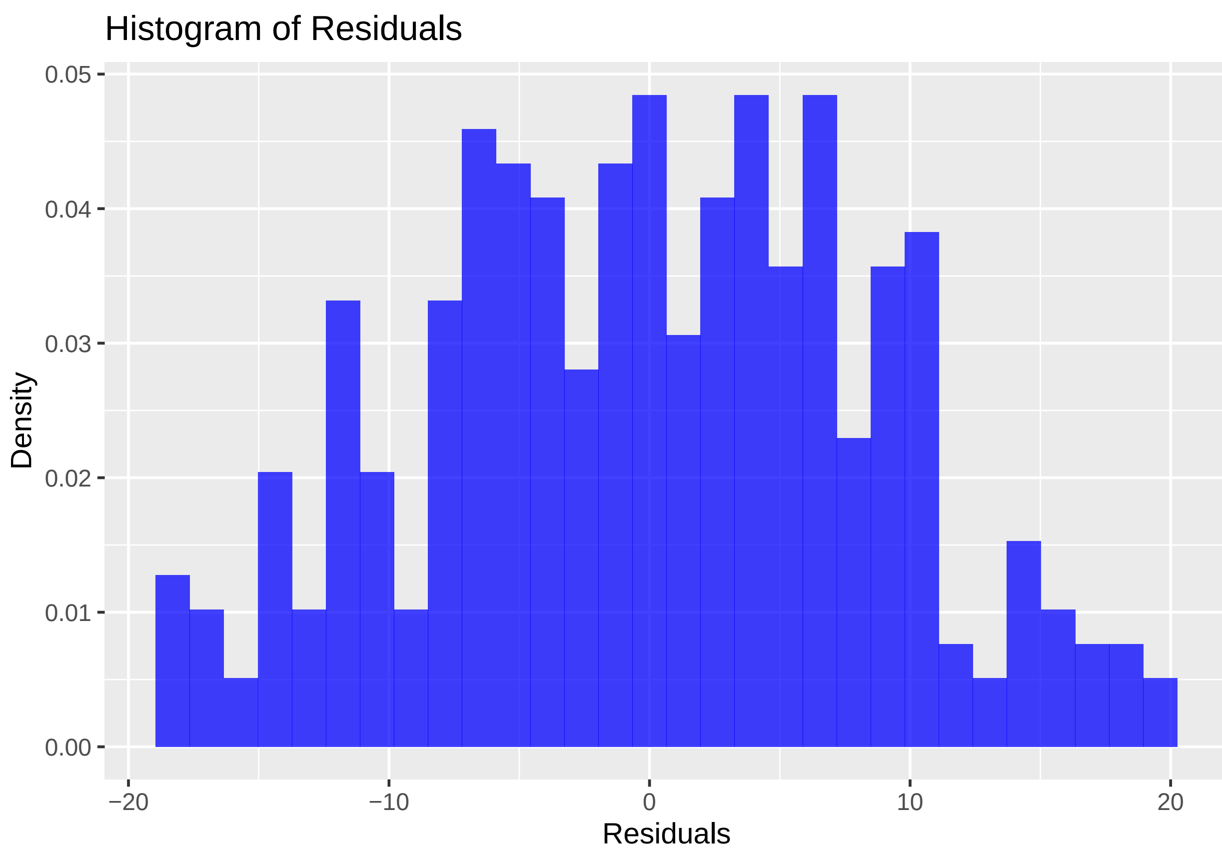
```r
### **Interpretation:** The estimated slope ($\beta_1$) is the second coefficient of the
### linear regression model, which is 0.000477127394928402. The estimated intercept is
### the first coefficient of the linear regression model, which is 59.5210944765007. When
### the Income is 0, the estimated Health_Score is about 59.521. For each additional 1 unit
### increase in income, Health_Score increases by about 0.000477.






##### HERE IS QUESTION 3 OF PART 2
# for simple linear regression: r_squared = r ^ 2
r_squared2 <- r_income ^ 2
r_squared2
```

```
## [1] 0.4472222
```

```r
### **Interpretation:** For simple linear regression models, r_squared = r ^ 2. Here, the
### R-squared value is 0.4472222.






##### HERE IS QUESTION 4 OF PART 2
# make a tibble with Predictor(Income) and Residuals
residuals_df <- tibble(
  # the first column is Income
  Predictor = health$Income,
  # the second column is Residuals
  Residuals = residuals(lm_model))
# draw the residual plot
residuals_df %>%
  # set the x-axis and y-axis aesthetic
  ggplot(aes(x = Predictor, y = Residuals)) +
  # set the degree of transparency
  geom_point(alpha = 0.75) +
  # draw the horizontal line
  geom_abline(intercept = 0, slope = 0, color = "red", linewidth = 1) +
  # set the title, x-axis and y-axis
  labs(title = "Residual Plot",
       x = "Predictor Values (Income)",
       y = "Residuals")
```

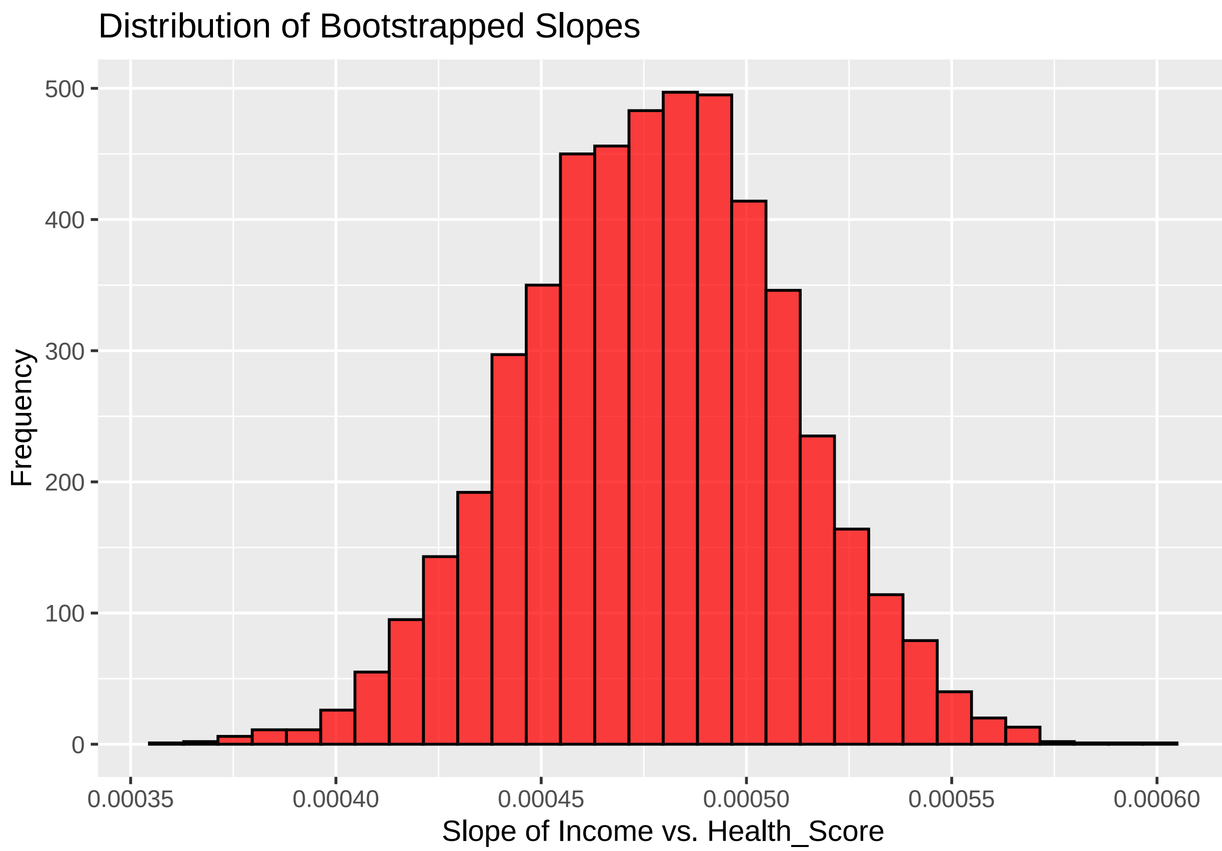



```r
# Draw the histogram to check the normality
residuals_df %>%
  # set the x-axis as Residuals
  ggplot(aes(x = Residuals)) +
  # draw the histogram and the y-axis is the probability density
  geom_histogram(aes(y = after_stat(density)), bins = 30, fill = "blue", alpha = 0.75) +
  # set the title, x-axis and y-axis
  labs(title = "Histogram of Residuals",
       x = "Residuals",
       y = "Density")
```




```
### **Interpretation and Reporting:** The first graph is the residual plot. The points all
### look evenly separated without any patterns. Based on the second graph, the residuals
### spread in normal distribution because the bars holistically form a bell shape, which is
### the same as normal distribution.






##### HERE IS QUESTION 1.1 OF PART 3
# set a random seed, so the result can be repeatedly shown
set.seed(123)
# 5000 bootstrap resamples means replicating 5000 times
boots_slopes <- replicate(5000, {
  # bootstrap resamples means we need to put the sample back, replace = TRUE
  sample_data <- health %>% sample_frac(size = 1, replace = TRUE)
  # model the linear regression
  model <- lm(Health_Score ~ Income, data = sample_data)
  # get the value of slope, which is the second one
  coef(model)[2]
})
# make a tibble of the 5000 slopes
boots_slopes_df <- tibble(slope = boots_slopes)
# based on the tibble of bootstrapped slopes, plot a histogram
boots_slopes_df %>%
```

```r
ggplot(aes(x = slope)) +
  # separate into 30 groups
  geom_histogram(bins = 30, fill = "red", color = "black", alpha = 0.75) +
  # name the title, x-axis and y-axis here
  labs(title = "Distribution of Bootstrapped Slopes",
       x = "Slope of Income vs. Health_Score",
       y = "Frequency")
```




```r
### **Interpretation:** The distribution of the bootstrapped slopes is shown here. It roughly
### forms a bell shape which makes it look like normal distribution.




##### HERE IS QUESTION 1.2 OF PART 3
# 95% CI means between 2.5% percentile and 97.5% percentile
confidence_interval3 <- quantile(boots_slopes, c(0.025, 0.975))
confidence_interval3
```

```
##         2.5%        97.5%
## 0.0004143684 0.0005404859
```

```r
### **Interpretation:** The 95% bootstrap confidence interval is (0.0004143684, 0.0005404859)
### according to the code result.
```

```r
##### HERE IS QUESTION 1.3 OF PART 3
### **Interpretation:** We are about 95% confident that the Health_Score tends to increase
### by between 0.0004143684 and 0.0005404859 units every one unit increase in Income on average.
```

```r
##### HERE IS QUESTION 2.1 OF PART 3
# make the new data point as a tibble
new_data <- tibble(Age = 22, BMI = 19, Exercise_Hours = 2, Income = 25000)
# predict the Health_Score
predicted_health_score <- predict(lm_model, newdata = new_data)
# show the predicted Health_Score
predicted_health_score
```

```
##        1
## 71.44928
```

```r
### **Interpretation:** The predicted Health_Score here is 71.44928 based on the linear
### regression model.
```

```r
##### HERE IS QUESTION 2.2 OF PART 3
# set a random seed, so the result can be repeatedly shown
set.seed(123)
# 5000 bootstrap resamples means replicating 5000 times
boots_predictions <- replicate(5000, {
  # bootstrap resamples means we need to put the sample back, replace = TRUE
  sample_data <- health %>% sample_frac(size = 1, replace = TRUE)
  # model the linear regression
  model <- lm(Health_Score ~ Income, data = sample_data)
  # return a vector of length 1
  predict(model, newdata = new_data)[1]
})
# 95% CI means between 2.5% percentile and 97.5% percentile
prediction_interval3 <- quantile(boots_predictions, c(0.025, 0.975))
prediction_interval3
```

```
##     2.5%    97.5%
## 69.49780 73.35643
```

```r
##### HERE IS QUESTION 2.3 OF PART 3
### **Interpretation:** According to the bootstrapping process, the 95% prediction interval
### for the response variable is (69.49780, 73.35643). The predict value is in the 95%
### prediction interval, which is acceptable.
```

```r
##### HERE IS QUESTION 1 OF PART 4
# set the 4 numerical predictors
predictors4 <- c("Age", "BMI", "Exercise_Hours", "Income")
# construct the multiple linear regression model
multi_lm_model <- lm(Health_Score ~ Age + BMI + Exercise_Hours + Income, data = health)
# get the coefficients of the multiple linear regression model
coefficients <- coef(multi_lm_model)
coefficients
```

```
##    (Intercept)            Age            BMI Exercise_Hours         Income
##  80.0935965699  -0.3065672205  -0.5565706448   1.4736346168   0.0004610996
```

```r
# get the value of the intercept, constant term
intercept <- coefficients[1]
intercept
```

```
## (Intercept)
##     80.0936
```

```r
# get the values of the slopes
slopes <- coefficients[-1]
# get the names of the corresponding variables of the slopes
predictor_names <- names(slopes)
multi_equation <- paste("Health_Score =", intercept, "+", paste(slopes, "*", predictor_names, collapse =
multi_equation
```

```
## [1] "Health_Score = 80.0935965698827 + -0.306567220507554 * Age + -0.556570644809524 * BMI + 1.47363
```

```r
### **Interpretation:** The equation of the multiple linear regression model comes from
### plugging out the coefficients. Based on the model, the equation is Health_Score =
### 80.0935965698827 + -0.306567220507554 * Age + -0.556570644809524 * BMI + 1.47363461678577
### * Exercise_Hours + 0.00046109957830263 * Income.





##### HERE IS QUESTION 2 OF PART 4
### **Interpretation:** The estimated intercept is 80.0935965699. When all the four
### variables are 0, the estimated Health_Score is about 80.094. The estimated slope
### coefficient is -0.306567220507554 (Age), -0.556570644809524 (BMI), 1.47363461678577
### (Exercise_Hours), and 0.00046109957830263 (Income). For each additional year of Age,
### Health_Score decreases by about 0.307 points. For each 1-unit increase in BMI, Health_Score
### decreases by about 0.557 points. For each additional hour of exercise, Health_Score
### increases by about 1.474 points. For each additional 1 unit increase in income,
### Health_Score increases by about 0.000461.
```

```r
##### HERE IS QUESTION 3 OF PART 4
# construct the simple linear model
simple_lm <- lm(Health_Score ~ Income, data = health)
# construct the multiple linear model
multi_lm <- lm(Health_Score ~ Age + BMI + Exercise_Hours + Income, data = health)
# get the adjusted R-squared with glance() for both the two cases
adj_r_squared_simple_lm <- simple_lm %>% glance() %>% pull(adj.r.squared)
adj_r_squared_multi_lm <- multi_lm %>% glance() %>% pull(adj.r.squared)
# make a tibble to show the adjusted R-squared values
model_comparison <- tibble(
  Model = c("Simple Linear Regression", "Multiple Linear Regression"),
  Adjusted_R2 = c(adj_r_squared_simple_lm, adj_r_squared_multi_lm)
)
# show the tibble
model_comparison
```

```
## # A tibble: 2 x 2
##   Model                      Adjusted_R2
##   <chr>                            <dbl>
## 1 Simple Linear Regression         0.445
## 2 Multiple Linear Regression       0.762
```

```r
### **Interpretation:** According to the comparison, the adjusted R-squared of the simple
### linear regression is 0.4453673 while that of the multiple linear regression is 0.7619176.
### The multiple linear regression model has a much larger adjusted R-squared.






##### HERE IS QUESTION 4 OF PART 4
# define the new observation according to the question
new_data4 <- tibble(Age = 22, BMI = 19, Exercise_Hours = 2, Income = 25000)

# Bootstrapping 95% Prediction Interval for Simple Regression
# set a random seed, so the result can be repeatedly shown
set.seed(123)
# 5000 bootstrap resamples means replicating 5000 times
boots_predictions_simple <- replicate(5000, {
  # bootstrap resamples means we need to put the sample back, replace = TRUE
  sample_data <- health %>% sample_frac(size = 1, replace = TRUE)
  # model the linear regression
  model <- lm(Health_Score ~ Income, data = sample_data)
  # return a vector of length 1
  predict(model, newdata = new_data4)[1]
})

# Bootstrapping 95% Prediction Interval for Multiple Regression
# set a random seed, so the result can be repeatedly shown
set.seed(123)
# 5000 bootstrap resamples means replicating 5000 times
```

```r
boots_predictions_multi <- replicate(5000, {
  # bootstrap resamples means we need to put the sample back, replace = TRUE
  sample_data <- health %>% sample_frac(size = 1, replace = TRUE)
  # model the linear regression
  model <- lm(Health_Score ~ Age + BMI + Exercise_Hours + Income, data = sample_data)
  # return a vector of length 1
  predict(model, newdata = new_data4)[1]
})

# Predict Health_Score with the simple regression model
prediction_simple4 <- predict(simple_lm, newdata = new_data4)
# Predict Health_Score with the multiple regression model
prediction_multi4 <- predict(multi_lm, newdata = new_data4)

# calculate the prediction interval of the simple regression model
prediction_interval_simple <- quantile(boots_predictions_simple, c(0.025, 0.975))
# calculate the prediction interval of the multiple regression model
prediction_interval_multi <- quantile(boots_predictions_multi, c(0.025, 0.975))

# Compare predictions in a tibble
prediction_comparison <- tibble(
  Model = c("Simple Regression", "Multiple Regression"),
  Predicted_Health_Score = c(prediction_simple4, prediction_multi4),
  `95% PI (Lower Bound)` = c(prediction_interval_simple[1], prediction_interval_multi[1]),
  `95% PI (Upper Bound)` = c(prediction_interval_simple[2], prediction_interval_multi[2])
)

# Show the tibble
prediction_comparison
```

```
## # A tibble: 2 x 4
##   Model     Predicted_Health_Score `95% PI (Lower Bound)` `95% PI (Upper Bound)`
##   <chr>                      <dbl>                  <dbl>                  <dbl>
## 1 Simple R~                   71.4                   69.5                   73.4
## 2 Multiple~                   77.2                   75.1                   79.3
```

```r
### **Interpretation:** According to the data analysis, the multiple linear regression
### predicted Health_Score is 77.24903 and its prediction interval is (75.06579, 79.32674).
### The simple linear regression model has a predicted Health_Score of 71.44928 and a
### prediction interval of (69.49780, 73.35643). Compared with the simple linear regression
### model, the multiple linear regression model has a larger predicted Health_Score and a
### larger prediction interval in values.







##### HERE IS QUESTION 5 OF PART 4
print("Model assumptions:")
```

```
## [1] "Model assumptions:"
```

```r
print("The relationship between predictors and Health_Score should be linear.")
```

```
## [1] "The relationship between predictors and Health_Score should be linear."
print("Residuals should not be correlated.")

## [1] "Residuals should not be correlated."
print("Residuals should have equal variance.")

## [1] "Residuals should have equal variance."
print("Residuals should follow a normal distribution.")

## [1] "Residuals should follow a normal distribution."
print("Predictors should not be highly correlated.")

## [1] "Predictors should not be highly correlated."
print("Concerns:")

## [1] "Concerns:"
print("If the residual plot shows a pattern, we need to check some other regression models.")

## [1] "If the residual plot shows a pattern, we need to check some other regression models."
print("If residuals are not normal, we need to remove outliers or using a robust regression model.")

## [1] "If residuals are not normal, we need to remove outliers or using a robust regression model."
```